A correction note on a flower-to-complete LP argument for rank-one Boolean tensor recovery

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Abstract

This note examines a proposed proof that recovery guarantees for the flower LP automatically imply recovery guarantees for the complete LP in the rank-one Boolean tensor factorization problem of Del Pia and Khajavirad. The main conclusion is negative for the sign-reduced complete LP displayed in their paper: the containment of the complete-LP feasible region in the flower-LP feasible region is false for that displayed formulation. In fact, a single false positive whose three indices all lie outside the planted factor supports is enough for the sign-reduced complete LP to fit the corrupted tensor exactly. Consequently, in the usual semi-random model, if all three factor complements have growing product size, an adversary can make the sign-reduced complete LP fail with high probability for every fixed positive corruption probability.

There is also a positive but limited statement. If one restores the deleted local convex-hull inequalities and works with an unreduced full complete LP, then the full complete LP does project into the flower LP. Therefore it inherits the known flower-LP recovery theorem. This inherited result is only partial progress: it gives no threshold beyond the flower-LP threshold and does not solve the sharper complete-LP recovery problem suggested by the numerical experiments in the original paper.

Status of the result. This note does *not* obtain a new positive recovery threshold for the complete LP. It gives (i) a corrected conditional inheritance theorem for a full or augmented complete LP, and (ii) a counterexample and a semi-random no-go result for the sign-reduced complete LP as displayed in the paper. Thus the uploaded flower-to-complete argument can be repaired only after changing the LP, or after weakening the conclusion to a conditional statement.

1 The three LPs

Let $[n] = \{1, \dots, n\}$, and similarly for $[m]$ and $[\ell]$. Let

$$G = (g_{ijk}) \in \{0,1\}^{n\times m\times \ell}$$

be the observed tensor, and define

$$S_0 := \{(i,j,k) : g_{ijk} = 0\}, \qquad S_1 := \{(i,j,k) : g_{ijk} = 1\}.$$

All LPs below minimize the same linear objective in the tensor-entry variables:

$$L_G(w) := \sum_{(i,j,k)\in S_0} w_{ijk} + \sum_{(i,j,k)\in S_1} (1 - w_{ijk}). \tag{1}$$

The planted tensor is

$$\overline{W} = \overline{x} \otimes \overline{y} \otimes \overline{z}, \qquad \overline{w}_{ijk} = \overline{x}_i \overline{y}_j \overline{z}_k.$$

Throughout, "recovery" means uniqueness after projection to the recovery variables (x, y, z, w). Auxiliary pair-product variables in extended formulations are not part of the object being recovered unless explicitly stated.

1.1 The flower LP

The flower LP has variables (x, y, z, w), with $(x, y, z) \in [0, 1]^{n+m+\ell}$, and minimizes (1) subject to

$$\begin{aligned}
&w_{ijk} \le x_i, \qquad w_{ijk} \le y_j, \qquad w_{ijk} \le z_k, && \forall (i,j,k) \in S_1, && \text{(2a)}\\
&w_{ijk} \ge 0, \qquad w_{ijk} \ge x_i + y_j + z_k - 2, && \forall (i,j,k) \in S_0, && \text{(2b)}\\
&w_{i'jk} - w_{ijk} \le 1 - x_i, && \forall (i,j,k) \in S_0,\ (i',j,k) \in S_1, && \text{(2c)}\\
&w_{ij'k} - w_{ijk} \le 1 - y_j, && \forall (i,j,k) \in S_0,\ (i,j',k) \in S_1, && \text{(2d)}\\
&w_{ijk'} - w_{ijk} \le 1 - z_k, && \forall (i,j,k) \in S_0,\ (i,j,k') \in S_1. && \text{(2e)}
\end{aligned}$$

1.2 The sign-reduced complete LP

The complete LP displayed in Del Pia and Khajavirad is a sign-reduced formulation. It uses auxiliary variables

$$u_{ij} \approx x_i y_j, \qquad v_{ik} \approx x_i z_k, \qquad s_{jk} \approx y_j z_k,$$

with $(x, y, z, u, v, s) \in [0, 1]^{n+m+\ell+nm+n\ell+m\ell}$. It minimizes (1) subject to

$$\begin{aligned}
w_{ijk} \le u_{ij}, \qquad w_{ijk} \le v_{ik}, \qquad w_{ijk} \le s_{jk}, &\quad \forall (i,j,k) \in S_1, && \text{(3a)}\\
x_i + y_j + z_k - u_{ij} - v_{ik} - s_{jk} + w_{ijk} \le 1, &\quad \forall (i,j,k) \in S_1, && \text{(3b)}\\
u_{ij} + v_{ik} - w_{ijk} \le x_i, &\quad \forall (i,j,k) \in S_0, && \text{(3c)}\\
u_{ij} + s_{jk} - w_{ijk} \le y_j, &\quad \forall (i,j,k) \in S_0, && \text{(3d)}\\
v_{ik} + s_{jk} - w_{ijk} \le z_k, &\quad \forall (i,j,k) \in S_0, && \text{(3e)}\\
w_{ijk} \ge 0, &\quad \forall (i,j,k) \in S_0. && \text{(3f)}
\end{aligned}$$

We call this LP $\mathrm{cLP}_{\mathrm{red}}$. The subscript is important: $\mathrm{cLP}_{\mathrm{red}}$ is obtained by deleting, depending on the sign of the objective coefficient of w_{ijk}, some local convex-hull inequalities.

1.3 The unreduced full complete LP

For comparison, define $\mathrm{cLP}_{\mathrm{full}}$ by imposing all the local convex-hull inequalities appearing in (3) for every triple (i, j, k), independent of whether $(i, j, k) \in S_0$ or $(i, j, k) \in S_1$. Thus $\mathrm{cLP}_{\mathrm{full}}$ minimizes (1) subject to

$$\begin{aligned}
w_{ijk} \le u_{ij}, \qquad w_{ijk} \le v_{ik}, \qquad w_{ijk} \le s_{jk}, &\quad \forall i, j, k, && \text{(4a)}\\
x_i + y_j + z_k - u_{ij} - v_{ik} - s_{jk} + w_{ijk} \le 1, &\quad \forall i, j, k, && \text{(4b)}\\
u_{ij} + v_{ik} - w_{ijk} \le x_i, &\quad \forall i, j, k, && \text{(4c)}\\
u_{ij} + s_{jk} - w_{ijk} \le y_j, &\quad \forall i, j, k, && \text{(4d)}\\
v_{ik} + s_{jk} - w_{ijk} \le z_k, &\quad \forall i, j, k, && \text{(4e)}\\
w_{ijk} \ge 0, &\quad \forall i, j, k, && \text{(4f)}
\end{aligned}$$

again with $(x, y, z, u, v, s) \in [0, 1]^{n+m+\ell+nm+n\ell+m\ell}$. This is the natural unreduced local relaxation obtained before sign-based deletion.

2 A correct inheritance statement for the full LP

The monotonicity idea in the uploaded file is valid only for LPs whose feasible projections are contained in the flower-LP feasible region. The full LP $\mathrm{cLP}_{\mathrm{full}}$ has this property. The sign-reduced LP $\mathrm{cLP}_{\mathrm{red}}$ does not, as shown in Section 3.

Lemma 2.1 (Full local constraints imply the flower constraints)**.** *The projection of the feasible region of* $\mathrm{cLP}_{\mathrm{full}}$ *onto the variables* (x, y, z, w) *is contained in the feasible region of* fLP*.*

Proof. Fix a triple (i, j, k), and abbreviate

$$a = x_i, \quad b = y_j, \quad c = z_k, \quad p = u_{ij}, \quad q = v_{ik}, \quad r = s_{jk}, \quad t = w_{ijk}.$$

The full local constraints say

$$t \le p, q, r, \qquad a + b + c - p - q - r + t \le 1,$$

$$p + q - t \le a, \qquad p + r - t \le b, \qquad q + r - t \le c, \qquad t \ge 0.$$

First, $t \le q$ and $p + q - t \le a$ imply $p \le a$. Similarly, $p \le b$, $q \le a$, $q \le c$, $r \le b$, and $r \le c$. Hence

$$t \le p \le a, b, \qquad t \le q \le a, c, \qquad t \le r \le b, c.$$

In particular, if $(i, j, k) \in S_1$, then $w_{ijk} \le x_i, y_j, z_k$, which gives (2a).

Next, the inequality $a + b + c - p - q - r + t \le 1$ gives

$$p + q + r \ge a + b + c + t - 1.$$

Adding the three running-intersection inequalities gives

$$2(p + q + r) - 3t \le a + b + c.$$

Combining the last two displays yields

$$2(a + b + c + t - 1) - 3t \le a + b + c,$$

which is equivalent to $t \ge a + b + c - 2$. Together with $t \ge 0$, this gives (2b) for triples in S_0.

It remains to derive the flower inequalities. Add the main inequality to the two running-intersection inequalities that contain r:

$$\begin{aligned}&(a + b + c - p - q - r + t) + (p + r - t - b) + (q + r - t - c)\\ &\qquad \le 1.\end{aligned}$$

After cancellation this gives

$$a + r - t \le 1, \qquad \text{or equivalently} \qquad r - t \le 1 - a. \tag{5}$$

Similarly, adding the main inequality to the two running-intersection inequalities that contain q gives

$$q - t \le 1 - b,$$

and adding the main inequality to the two running-intersection inequalities that contain p gives

$$p - t \le 1 - c.$$

Now suppose $(i,j,k) \in S_0$ and $(i',j,k) \in S_1$. Applying (4a) to the triple (i',j,k) gives $w_{i'jk} \le s_{jk} = r$, while (5) applied to the triple (i,j,k) gives $s_{jk} - w_{ijk} \le 1 - x_i$. Hence

$$w_{i'jk} - w_{ijk} \le 1 - x_i.$$

This is (2c). The proofs of (2d) and (2e) are identical, using the local consequences $q - t \le 1 - y_j$ and $p - t \le 1 - z_k$, respectively. $\square$

Proposition 2.2 (Deterministic inheritance)**.** *Let $\mathcal{Q}(G)$ be the feasible region of any LP relaxation with objective (1). Suppose that the planted point $(\overline{x}, \overline{y}, \overline{z}, \overline{w})$ has a feasible lift to $\mathcal{Q}(G)$, and that the projection of $\mathcal{Q}(G)$ to (x,y,z,w) is contained in the feasible region of* fLP*. If* fLP *uniquely recovers $(\overline{x}, \overline{y}, \overline{z}, \overline{w})$, then the LP over $\mathcal{Q}(G)$ also recovers $(\overline{x}, \overline{y}, \overline{z}, \overline{w})$ after projection.*

Proof. Let $q^* = (\overline{x}, \overline{y}, \overline{z}, \overline{w})$, and let $Q \in \mathcal{Q}(G)$ be an optimal solution. Let q be its projection to (x,y,z,w). By the projection assumption, q is feasible for fLP. Therefore flower optimality gives

$$L_G(q) \ge L_G(q^*).$$

Since q^* has a feasible lift $Q^* \in \mathcal{Q}(G)$ and Q is optimal for $\mathcal{Q}(G)$,

$$L_G(q) = L_G(Q) \le L_G(Q^*) = L_G(q^*).$$

Thus $L_G(q) = L_G(q^*)$. Since the flower LP has q^* as its unique optimum, $q = q^*$. This proves recovery after projection. $\square$

Corollary 2.3 (Inherited flower-LP threshold for the full LP)**.** *The full LP* $\mathrm{cLP}_{\mathrm{full}}$ *inherits every deterministic and probabilistic recovery guarantee proved for* fLP*. In particular, under the semi-random hypotheses of the flower-LP theorem of Del Pia and Khajavirad,* $\mathrm{cLP}_{\mathrm{full}}$ *recovers with high probability whenever*

$$p < \frac{r_{\overline{W}}}{1 + 2r_{\overline{W}}}.$$

Proof. This follows immediately from Lemma 2.1, Proposition 2.2, and the flower-LP recovery theorem. $\square$

Remark 2.4. *Corollary 2.3 is a valid repair of the uploaded proof only if the LP is the unreduced full LP, or if the missing full local constraints are explicitly added back. It is not a new solution of the complete-LP open problem: it gives exactly the flower-LP threshold and does not use the additional strength of the complete LP to improve that threshold.*

3 Failure of the sign-reduced complete LP

The sign-reduced LP $\mathrm{cLP}_{\mathrm{red}}$ does not project into the flower LP. The failure is not cosmetic; it leads to genuine recovery failures.

Lemma 3.1 (Minimal containment failure)**.** *For $n = m = \ell = 1$ and $G_{111} = 1$, the point*

$$w_{111} = u_{11} = v_{11} = s_{11} = 1, \qquad x_1 = y_1 = z_1 = 0$$

is feasible for $\mathrm{cLP}_{\mathrm{red}}$*, but its projection is infeasible for* fLP*.*

Proof. Since $S_1 = \{(1,1,1)\}$ and $S_0 = \emptyset$, the sign-reduced complete LP imposes

$$w_{111} \le u_{11}, \quad w_{111} \le v_{11}, \quad w_{111} \le s_{11},$$

and

$$x_1 + y_1 + z_1 - u_{11} - v_{11} - s_{11} + w_{111} \le 1.$$

The displayed point satisfies these inequalities. However, fLP requires $w_{111} \le x_1$, $w_{111} \le y_1$, and $w_{111} \le z_1$, which are violated. □

3.1 A general one-false-positive obstruction

The following proposition is the main reason the sign-reduced formulation cannot have the same semi-random recovery theorem as the flower LP.

Proposition 3.2 (One outside false positive is fitted exactly by $\mathrm{cLP}_{\mathrm{red}}$)**.** *Let*

$$\overline{W} = \mathbf{1}_A \otimes \mathbf{1}_B \otimes \mathbf{1}_C$$

for sets $A \subseteq [n]$, $B \subseteq [m]$, and $C \subseteq [\ell]$. Suppose that $i_0 \notin A$, $j_0 \notin B$, and $k_0 \notin C$. Let the observed tensor G be obtained from $\overline{W}$ by adding the single false positive (i_0, j_0, k_0):

$$S_1 = A \times B \times C \ \cup \ \{(i_0, j_0, k_0)\}.$$

Then $\mathrm{cLP}_{\mathrm{red}}$ has optimal value zero. In particular, the planted tensor, whose objective value is one, is not optimal.

Proof. Set $w_{ijk} = g_{ijk}$ for every triple, and set all factor variables equal to one:

$$x_i = y_j = z_k = 1 \qquad \text{for all } i, j, k.$$

Set the pair variables to be the pairwise projections of S_1:

$$u_{ij} = 1 \iff (i,j) \in (A \times B) \cup \{(i_0, j_0)\},$$

$$v_{ik} = 1 \iff (i,k) \in (A \times C) \cup \{(i_0, k_0)\}, \qquad s_{jk} = 1 \iff (j,k) \in (B \times C) \cup \{(j_0, k_0)\}.$$

All remaining pair variables are set to zero.

For every triple in S_1, all three of its pair projections are equal to one. Hence (3a) holds, and

$$x_i + y_j + z_k - u_{ij} - v_{ik} - s_{jk} + w_{ijk} = 3 - 1 - 1 - 1 + 1 = 1,$$

so (3b) holds as equality.

Now consider a triple $(i,j,k) \in S_0$. We claim that at most one of u_{ij}, v_{ik}, s_{jk} is equal to one. Indeed, if two of these pair variables were one, then either the two pair memberships would both come from $A \times B$, $A \times C$, $B \times C$, forcing $(i,j,k) \in A \times B \times C$, or they would both come from the exceptional pairs, forcing $(i,j,k) = (i_0, j_0, k_0)$, or they would give an impossible equality such as $i \in A$ and $i = i_0 \notin A$. Thus (i,j,k) would not be in S_0, a contradiction. Therefore, for $(i,j,k) \in S_0$,

$$u_{ij} + v_{ik} \le 1 = x_i, \qquad u_{ij} + s_{jk} \le 1 = y_j, \qquad v_{ik} + s_{jk} \le 1 = z_k,$$

because $w_{ijk} = 0$. This verifies (3c)–(3e); (3f) is immediate.

The constructed feasible point has $w_{ijk} = 1$ on S_1 and $w_{ijk} = 0$ on S_0, so its objective value is zero. This is the minimum possible value, since $1 - w_{ijk} \ge 0$ on S_1 by (3a) and the upper bound on the pair variables, while $w_{ijk} \ge 0$ on S_0. The planted tensor has objective value one because it disagrees with G exactly at (i_0, j_0, k_0). Hence the planted tensor is not optimal for $\mathrm{cLP}_{\mathrm{red}}$. □

Corollary 3.3 (Semi-random no-go for the sign-reduced LP)**.** *Consider a sequence of planted tensors*

$$\overline{W}^{(n,m,\ell)} = \mathbf{1}_{A_n} \otimes \mathbf{1}_{B_m} \otimes \mathbf{1}_{C_\ell}.$$

Let

$$M_{n,m,\ell} := |[n] \setminus A_n|\,|[m] \setminus B_m|\,|[\ell] \setminus C_\ell|.$$

If the fully-random corruption probability $p = p_{n,m,\ell}$ *satisfies*

$$p_{n,m,\ell} M_{n,m,\ell} \longrightarrow \infty,$$

then, with probability tending to one, there is a monotone semi-random adversary for which $\mathrm{cLP}_{\mathrm{red}}$ *fails to recover the planted tensor. In particular, if* p *is a fixed positive constant and the three complements have product size tending to infinity,* $\mathrm{cLP}_{\mathrm{red}}$ *has no high-probability semi-random recovery guarantee.*

Proof. In the fully-random corruption step, every entry in

$$([n] \setminus A_n) \times ([m] \setminus B_m) \times ([\ell] \setminus C_\ell)$$

is a planted zero. The probability that none of these $M_{n,m,\ell}$ entries is flipped to one is

$$(1 - p_{n,m,\ell})^{M_{n,m,\ell}} \le \exp(-p_{n,m,\ell} M_{n,m,\ell}) \to 0.$$

Thus with high probability at least one such false positive exists. A monotone adversary may revert every other corrupted entry back to its planted value and leave this one false positive unchanged. The resulting tensor has exactly the form in Proposition 3.2, so $\mathrm{cLP}_{\mathrm{red}}$ fails. □

3.2 A finite instance where the flower LP succeeds but $\mathrm{cLP}_{\mathrm{red}}$ fails

The next proposition shows that the failure of $\mathrm{cLP}_{\mathrm{red}}$ is not merely a case where the flower LP also fails. Even the flower LP, and hence the full complete LP, can recover while the sign-reduced complete LP fits the corrupted tensor exactly.

Proposition 3.4 (Flower recovery versus sign-reduced complete-LP failure)**.** *Let* $n = m = \ell = 3$*, let*

$$A = B = C = \{1, 2\}, \qquad \overline{W} = \mathbf{1}_A \otimes \mathbf{1}_B \otimes \mathbf{1}_C,$$

and let G *be obtained from* $\overline{W}$ *by adding the single false positive* $(3, 3, 3)$*. Then:*

(i) fLP *uniquely recovers* $(\mathbf{1}_A, \mathbf{1}_B, \mathbf{1}_C, \overline{W})$*, with optimal value one;*

(ii) $\mathrm{cLP}_{\mathrm{full}}$ *recovers the same projected variables;*

(iii) $\mathrm{cLP}_{\mathrm{red}}$ *has optimal value zero and therefore does not recover* $\overline{W}$*.*

Proof. The claim for $\mathrm{cLP}_{\mathrm{red}}$ is exactly Proposition 3.2 with $(i_0, j_0, k_0) = (3, 3, 3)$. The claim for $\mathrm{cLP}_{\mathrm{full}}$ follows from the flower-LP claim and Proposition 2.2. It remains to prove the flower-LP claim.

For the planted solution, the objective value is one, because the only disagreement between G and $\overline{W}$ is the false positive $(3, 3, 3)$. We show that every flower-LP feasible point has objective at least one, with equality only at the planted recovery variables.

Let (x, y, z, w) be feasible for fLP, and write

$$a := w_{333}, \qquad b := w_{111}, \qquad u := w_{311}.$$

Since $(3, 3, 3) \in S_1$, the upper bound (2a) gives $a \le x_3$. Since $(3, 1, 1) \in S_0$ and $(1, 1, 1) \in S_1$, the flower inequality (2c) gives

$$b - u \le 1 - x_3.$$

Consequently,

$$a \le 1 - b + u. \tag{6}$$

All terms in the flower-LP objective are nonnegative: on S_1, (2a) and $(x, y, z) \in [0, 1]^{n+m+\ell}$ imply $w_{ijk} \le 1$, and on S_0, (2b) implies $w_{ijk} \ge 0$. Hence

$$L_G(w) \ge (1 - a) + (1 - b) + u \ge (1 - a) + a = 1,$$

where the second inequality is (6). Thus the flower-LP optimum is exactly one.

Now suppose $L_G(w) = 1$. Equality in the preceding display forces all objective terms other than $(1 - a)$, $(1 - b)$, and u to be zero, and also forces equality in (6). In particular,

$$w_{112} = 1, \qquad w_{312} = 0.$$

Applying (2c) to $(3, 1, 2) \in S_0$ and $(1, 1, 2) \in S_1$ gives

$$1 - 0 = w_{112} - w_{312} \le 1 - x_3,$$

so $x_3 = 0$. Since $a \le x_3$, we get $a = 0$. Equality in (6) then gives $1 - b + u = 0$, and since $b \le 1$ and $u \ge 0$, this implies $b = 1$ and $u = 0$. Together with the already-zero remaining objective terms, this proves

$$w_{ijk} = 1 \quad \text{for } (i, j, k) \in A \times B \times C, \qquad w_{ijk} = 0 \quad \text{otherwise}.$$

Thus $w = \overline{W}$.

Finally, the block entries $w_{ijk} = 1$ for $i, j, k \in \{1, 2\}$, together with (2a), force

$$x_1 = x_2 = y_1 = y_2 = z_1 = z_2 = 1.$$

The zero entries $w_{311} = w_{131} = w_{113} = 0$, together with the lower bounds (2b), give

$$0 = w_{311} \ge x_3 + y_1 + z_1 - 2 = x_3,$$

$$0 = w_{131} \ge x_1 + y_3 + z_1 - 2 = y_3, \qquad 0 = w_{113} \ge x_1 + y_1 + z_3 - 2 = z_3.$$

Therefore $x_3 = y_3 = z_3 = 0$. The optimal projected solution is uniquely $(\mathbf{1}_A, \mathbf{1}_B, \mathbf{1}_C, \overline{W})$. $\square$

4 Consequences

The attempted flower-to-complete proof has two different fates, depending on which LP is meant.

(1) For the full local complete LP $\mathrm{cLP}_{\mathrm{full}}$, the proof is correct after the missing local inequalities are restored. The result is the inherited flower-LP guarantee in Corollary 2.3. This is only partial progress and is not a sharper complete-LP theorem.

(2) For the sign-reduced complete LP $\mathrm{cLP}_{\mathrm{red}}$ displayed in the paper, the proof is not repairable by a containment argument: Lemma 3.1 shows containment fails, Proposition 3.4 shows that fLP can recover while $\mathrm{cLP}_{\mathrm{red}}$ fails, and Corollary 3.3 shows that the usual semi-random guarantee is impossible for $\mathrm{cLP}_{\mathrm{red}}$ whenever all three factor complements have growing product size.

Thus this note should be viewed as a correction and a negative/clarifying partial result, not as a solution that proves the conjectured superior recovery behavior of the complete LP. The fully-random behavior of the sign-reduced LP, and positive recovery thresholds that genuinely exploit complete-LP constraints beyond the flower inequalities, remain separate problems.

References

[1] Alberto Del Pia and Aida Khajavirad. Rank-One Boolean Tensor Factorization and the Multilinear Polytope. *Mathematics of Operations Research*, 50(2):1514–1554, 2025. DOI: 10.1287/moor.2022.0201.

[2] Alberto Del Pia and Aida Khajavirad. Rank-one Boolean tensor factorization and the multilinear polytope. arXiv:2202.07053v3, 2024. Available at `https://arxiv.org/abs/2202.07053`.